C **Foundations of Programming** Lab Assignments 16 – 22

A#16 – Even/Odd | A#17 – Sine Series | A#18 – Strings | A#19 – Employee | A#20 – Point

ASSIGNMENT NO. 16

Partition List into Even and Odd Numbers

Accept N integers and split them into two sublists — even and odd — using the modulus operator.

C CODE

```c
#include <stdio.h>

int main() {
    int n, i;
    printf("Enter number of elements: ");
    scanf("%d", &n);

    int arr[n], even[n], odd[n];
    int ec = 0, oc = 0;

    printf("Enter %d integers: ", n);
    for (i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    for (i = 0; i < n; i++) {
        if (arr[i] % 2 == 0)
            even[ec++] = arr[i];
        else
            odd[oc++] = arr[i];
    }

    printf("\nEven numbers are: ");
    for (i = 0; i < ec; i++) printf("%d ", even[i]);

    printf("\nOdd numbers are: ");
    for (i = 0; i < oc; i++) printf("%d ", odd[i]);

    printf("\n");
    return 0;
}
```

SAMPLE I/O

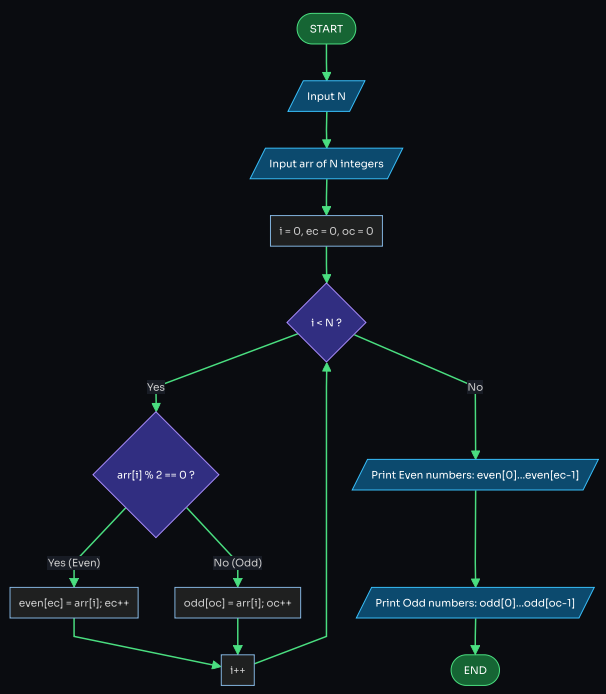
```
Input:
Enter number of elements: 5
Enter 5 integers: 10 7 4 9 12

Output:
Even numbers are: 10 4 12
Odd numbers are: 7 9
```

KEY CONCEPTS

OPERATOR
% returns remainder of division

CONDITION
arr[i]%2==0 → Even, else Odd

COUNTERS
ec = even count, oc = odd count

ARRAYS
Separate even[] and odd[] storage

FLOWCHART